

NESBYEN-TODOKK, Norway

WMO: 013730

Lat: 60.567N

Lon: 9.133E

Elev: 545

StdP: 14.41

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-4.9	-13.3	2.7	-9.1	-8.5	3.5	-4.6	12.4	37.1	10.1	36.5	0.4	0	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	81.6	60.8	77.5	59.2	73.6	58.0	63.4	74.0	61.8	71.7	60.3	69.6	5.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.1	79.1	65.4	58.4	74.4	64.4	56.6	69.7	62.5	29.0	73.8	27.9	71.3	26.8	69.8	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.0	8.5	-11.2	86.1	5.9	3.7	-15.5	88.8	-19.0	90.9	-22.3	93.0	-26.6	95.7		
(5)			-11.5	66.1	5.6	2.1	-15.6	67.6	-18.8	68.9	-22.0	70.1	-26.1	71.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	17.2	21.2	31.1	40.6	49.3	57.3	62.0	58.0	50.6	38.6	27.9	19.7		
	DBStd	17.30	11.52	10.76	8.59	5.55	6.95	5.23	5.13	4.37	5.47	7.24	8.51	12.16		
	HDD50	4826	1017	807	587	285	99	5	0	3	59	359	665	940		
	HDD65	9331	1482	1227	1051	732	488	238	124	221	431	818	1114	1405		
	CDD50	1009	0	0	1	3	77	224	371	249	78	6	0	0		
	CDD65	40	0	0	0	0	1	6	31	2	0	0	0	0		
	CDH74	759	0	0	0	0	58	215	442	42	2	0	0	0		
	CDH80	157	0	0	0	0	7	43	105	2	0	0	0	0		
	WSAvg	2.7	1.9	2.0	3.2	3.6	3.6	3.7	3.2	2.6	2.6	2.1	1.8	1.8		
	PrecAvg	21.9	1.5	0.9	0.9	1.1	1.7	2.6	3.3	3.1	2.1	2.0	1.7	1.2		
(15) Precipitation	PrecMax	29.1	2.5	2.0	2.2	2.4	5.4	5.6	7.3	6.6	4.7	5.5	6.1	2.2		
	PrecMin	15.4	0.4	0.1	0.0	0.1	0.0	0.2	0.8	0.6	0.7	0.2	0.3	0.2		
	PrecStd	2.9	0.6	0.5	0.5	0.7	1.1	1.3	1.5	1.4	0.8	1.2	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.2	49.3	58.4	67.5	80.5	85.1	86.8	78.8	72.4	63.8	52.6	48.6		
		MCWB	40.6	41.3	45.6	50.2	58.4	60.0	62.2	60.3	58.4	55.4	46.2	41.2		
	2%	DB	39.7	43.0	52.7	61.7	74.0	79.9	83.1	74.2	67.8	56.5	47.0	43.2		
		MCWB	34.7	36.6	42.5	47.5	55.1	58.6	62.4	60.6	56.0	48.8	42.8	37.6		
	5%	DB	35.6	38.8	48.3	57.8	69.6	75.6	79.3	70.8	64.3	53.5	42.4	39.5		
		MCWB	33.1	34.6	39.5	45.2	53.5	56.5	61.0	58.8	54.4	47.8	39.6	35.4		
	10%	DB	32.3	35.1	44.2	53.6	64.6	71.0	75.1	67.6	61.3	50.4	38.2	35.0		
		MCWB	30.9	32.3	36.6	42.8	52.0	54.6	59.5	57.0	53.0	45.4	36.7	32.9		
	0.4%	WB	40.9	41.3	46.8	51.7	60.9	62.1	66.3	64.3	60.1	55.6	47.5	41.9		
		MCDB	46.9	48.3	58.4	62.3	74.0	73.4	77.9	72.7	67.5	62.4	51.2	47.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	37.2	43.2	48.6	57.7	60.5	64.3	62.3	57.9	50.9	43.5	38.3		
		MCDB	38.9	42.5	52.2	59.7	69.6	74.7	76.2	69.9	64.9	54.4	46.5	42.6		
	5%	WB	33.4	34.9	40.3	46.0	55.1	58.6	62.9	60.7	56.1	48.8	39.7	35.4		
		MCDB	35.5	38.3	47.9	55.6	65.8	70.4	74.0	67.6	61.9	52.6	42.0	38.8		
	10%	WB	30.9	32.7	37.1	43.9	52.6	57.0	61.4	59.1	54.3	45.9	36.6	32.7		
		MCDB	32.6	35.2	43.3	52.5	62.3	67.4	71.3	65.5	59.4	49.8	38.3	34.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	14.4	19.4	21.7	22.8	22.0	20.6	18.6	19.2	13.9	9.7	11.4		
		MCDBR	16.1	18.6	24.3	30.0	31.8	32.5	29.4	23.7	24.3	18.6	14.6	18.2		
		MCWBR	12.8	13.1	14.9	16.8	15.7	14.4	11.8	11.5	14.3	12.1	11.2	13.3		
	5% WB	MCDBR	15.6	16.9	23.9	27.1	26.3	24.4	23.1	19.4	20.8	15.6	13.1	17.3		
		MCWBR	12.7	12.2	15.0	15.7	13.4	11.8	10.1	10.4	12.9	11.1	10.6	13.0		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.275	0.311	0.353	0.358	0.352	0.369	0.361	0.325	0.308	0.250	0.201		
	taud		2.156	2.248	2.227	2.240	2.390	2.450	2.418	2.432	2.517	2.466	2.236	2.051		
	Ebn at Noon		173	230	257	266	275	279	270	263	255	219	162	130		
	Edh at Noon		12	21	30	36	33	32	32	30	23	17	11	8		
(44) All-Sky Solar Radiation	RadAvg		101	312	787	1314	1520	1637	1470	1158	763	357	126	60		
	RadStd		18	68	65	174	168	157	183	101	98	64	27	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page